

OS Practical

Practical1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
lahari@DESKTOP-AIC33F7:~$ mkdir practical1
mkdir: cannot create directory 'practical1': File exists
lahari@DESKTOP-AIC33F7:~$ cd practical1
lahari@DESKTOP-AIC33F7:~/practical1$ nano practical1.c
lahari@DESKTOP-AIC33F7:~/practical1$ gcc practical1.c -o practical1
lahari@DESKTOP-AIC33F7:~/practical1$ ./practical1
```

Uname

```
Enter Linux command: uname

Parent Process
Parent PID : 1662

Child Process
Child PID : 1663
Linux
Child process completed.
lahari@DESKTOP-AIC33F7:~/practical1$ ./practical1
```

Lscpu

```
lanari@DESKTOP-AIC33F7:~/practical1$ ./practical1
Enter Linux command: lscpu
```

```
Parent Process
Parent PID : 1664
```

```
Child Process
Child PID : 1665
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 8
On-line CPU(s) list:    0-7
Vendor ID:              GenuineIntel
Model name:             11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz
CPU family:             6
Model:                  140
Thread(s) per core:     2
Core(s) per socket:     4
Socket(s):              1
Stepping:               1
BogoMIPS:               3379.19
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
                        e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology
                        tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid s
                        se4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor la
                        hf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsb
                        ase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma
                        clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves vnmi avx5
                        12vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rd
                        pid movdiri movdir64b fsrm avx512_vp2intersect md_clear ibt flush_lld arch_capabilities
```

```
Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                     192 KiB (4 instances)
L1i:                     128 KiB (4 instances)
L2:                      5 MiB (4 instances)
L3:                      12 MiB (1 instance)
NUMA:
NUMA node(s):            1
NUMA node0 CPU(s):       0-7
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit:          Not affected
L1tf:                    Not affected
Mds:                      Not affected
Meltdown:                Not affected
Mmio stale data:         Not affected
Old microcode:           Not affected
Reg file data sampling:  Not affected
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRsB-eIBRS SW sequence; BHI SW lo
                        op, KVM SW loop
Srbds:                   Not affected
Tsa:                     Not affected
Tsx async abort:         Not affected
```

Lsblk

```

Enter Linux command: lsblk

Parent Process

Child Process
Parent PID : 1667
Child PID : 1689
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda  8:0    0 356.9M  1 disk
sdb  8:16   0 159.4M  1 disk
sdc  8:32   0    2G    0 disk [SWAP]
sdd  8:48   0    1T    0 disk /mnt/wslg/distro
/

Child process completed.

```

Ps

```

lahari@DESKTOP-AIC33F7:~/practical1$ ps
  PID TTY          TIME CMD
 1631 pts/2        00:00:00 bash
 1692 pts/2        00:00:00 ps
lahari@DESKTOP-AIC33F7:~/practical1$ ./practical1
Enter Linux command: ps

Parent Process
Parent PID : 1693

Child Process
Child PID : 1694
  PID TTY          TIME CMD
 1631 pts/2        00:00:00 bash
 1693 pts/2        00:00:00 practical1
 1694 pts/2        00:00:00 ps
Child process completed.
lahari@DESKTOP-AIC33F7:~/practical1$ ./practical1

```

Top

```
lahari@DESKTOP-AIC33F7:~/practical1$ ./practical1
```

```
Enter Linux command: top
```

```
Parent Process
```

```
Parent PID : 1695
```

```
Child Process
```

```
Child PID : 1696
```

```
top - 05:12:58 up 31 min, 1 user, load average: 0.00, 0.00, 0.00
```

```
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
```

```
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

```
MiB Mem : 7643.6 total, 7022.7 free, 510.7 used, 265.9 buff/cache
```

```
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7132.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
192	syslog	20	0	222516	6036	4652	S	0.3	0.1	0:00.06	rsyslogd
1	root	20	0	21672	12488	9148	S	0.0	0.2	0:00.70	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50444	16356	15256	S	0.0	0.2	0:00.20	systemd-journal
99	root	20	0	25284	6712	5208	S	0.0	0.1	0:00.27	systemd-udev
129	systemd+	20	0	21344	13244	11120	S	0.0	0.2	0:00.11	systemd-resolve
131	systemd+	20	0	91036	8044	7080	S	0.0	0.1	0:00.10	systemd-timesyn
167	root	20	0	4244	2744	2496	S	0.0	0.0	0:00.01	cron
168	message+	20	0	9680	5508	4764	S	0.0	0.1	0:00.11	dbus-daemon
182	root	20	0	17980	8944	7900	S	0.0	0.1	0:00.08	systemd-logind
198	root	20	0	3124	2040	1900	S	0.0	0.0	0:00.00	agetty
219	root	20	0	107036	23236	13784	S	0.0	0.3	0:00.08	unattended-upgr
307	root	20	0	6700	4760	3980	S	0.0	0.1	0:00.00	login
351	lahari	20	0	20324	11604	9480	S	0.0	0.1	0:00.08	systemd
352	lahari	20	0	21168	3576	1848	S	0.0	0.0	0:00.00	(sd-pam)
378	lahari	20	0	6080	5356	3692	S	0.0	0.1	0:00.01	bash
1410	polkitd	20	0	308172	7300	6404	S	0.0	0.1	0:00.06	polkitd
1501	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
1502	root	20	0	3200	1132	980	S	0.0	0.0	0:00.00	Relay(1507)
1507	lahari	20	0	6080	5388	3672	S	0.0	0.1	0:00.04	bash
1629	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
1630	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.04	Relay(1631)